Exercise 7.3.1 Consider the question why we tried a power-series solution for Eq. (7.3.11) but not Eq. (7.3.8). By feeding in a series into the latter, verify that a three-term recursion relation between C_{n+2}, C_n, and C_{n-2} obtains, from which the solution does not follow so readily. The problem is that ψ'' has two powers of y less than $2\epsilon\psi$, while the $-y^2$ piece has two more powers of y. In Eq. (7.3.11) on the other hand, of the three pieces u'', $-2yu'$, and $(2\epsilon-1)u$, the last two have the same powers of y.

Consider the equation,

$$\psi'' + (2\epsilon - y^2)\psi = 0 \tag{7.3.8}$$

,

And assume ψ is of form $\psi = \sum_0^\infty c_n y^n$. Then it becomes

$$= \sum_0^\infty c_n(n-1)y^{n-2} + c_n(2\epsilon - y^2)y^n = 0$$
$$= \sum_0^\infty c_{n+2}(n+2)(n-1)y^n + 2\epsilon y^n - c_n y^{n+2} = 0$$

In the previous case, we were able to arrive at a recursion relation because we were able to factor out and cancel a common y^n among all the terms, there by letting us express coefficients in terms of other coefficients only. Now we have a mixed order equation where some terms have order y^{n+2} and some terms have order y^n, thus the same approach does not follow algebraically and we cannot solve for a three-term recursion relation.

Exercise 7.3.2 Verify that $H_3(y)$ and $H_4(y)$ obey the recursion relation, Eq. (7.3.15).

$H_3(y) = -12(y - 2/3y^3)$. Since n is odd (3), we can set our C_0 to zero, fixing our even coefficients to zero by the recursion relation. Using $n = 3$, we also have our quantized energy $\epsilon_3 = 7/2$. Then the recursion relation will assume the form

$$C_3 = C_1\frac{2(1)+1-2\epsilon_3}{(1+2)(1+1)} = C1\frac{-4}{6}$$

This relation is satisfied by the implied coefficients $C_1 = -12$ and $C_3 = 8$ from the Hermite polynomial.

$H_4(y) = 12(1 - 4y^2 + 4/3y^4)$, with $\epsilon = 9/2$, again the relation specifies for the non-zero coefficients

$$C_2 = C_0\frac{2(0)+1-2\epsilon_4}{(0+2)(0+1)} = -4C_0$$
$$C_4 = C_2\frac{2(2)+1-2\epsilon_4}{(2+2)(2+1)} = C_2\frac{-4}{12}$$

Given the coefficients given by H_4, $C_0 = 12$, $C_2 = -48$, $C_4 = 16$, these relations are satisfied.

Exercise 7.3.3 If $\psi(x)$ is even and $\psi(x)$ is odd under $x \to -x$, show that

$$\int_{-\infty}^{\infty} \psi(x)\phi(x)dx = 0$$

Use this to show that $\psi_2(x)$ and $\psi_1(x)$ are orthogonal. Using the values of Gaussian integrals in Appendix A.2 verify that $\psi_2(x)$ and $\psi_1(x)$ are orthogonal.

By splitting the integral we can observe,

$$= \int_{-\infty}^{0} \psi(x)\phi(x)dx + \int_{0}^{\infty} \psi(x)\phi(x)dx$$

Note that $\psi(x) = \psi(-x)$, and $\phi(-x) = \phi(x)$, this will result in the two integrals being equal in magnitude but opposite in sign. To make it explicit, we can use the substitution $y = -x$, $dy = -dx$, and swap the first integral's limits (after substituting the limits).

$$\begin{aligned} &= -\int_{0}^{\infty} \psi(-y)\phi(-y)(-dy) + \int_{0}^{\infty} \psi(x)\phi(x)dx \\ &= -\int_{0}^{\infty} \psi(y)\phi(y)dy + \int_{0}^{\infty} \psi(x)\phi(x)dx = 0 \end{aligned}$$

Then for the solutions of the quantum harmonic oscillator, we have

$$\begin{aligned} \psi_1(x) &= \Big(\frac{m\omega}{\pi\hbar 4}\Big)^{1/4} \exp\Big(-\frac{m\omega x^2}{2\hbar}\Big)\Big(2x\big(\frac{m\omega}{\hbar}\big)^{1/2}\Big) \\ \psi_2(x) &= \Big(\frac{m\omega}{\pi\hbar 16 * 4}\Big)^{1/4} \exp\Big(-\frac{m\omega x^2}{2\hbar}\Big)\Big(-2\big(1 - 2x^2 m\omega/\hbar\big)\Big) \end{aligned}$$

Note that $\psi_1(x)$ is an odd function and $\psi_2(x)$ is an even function. Thus using the previously proven statement, these two functions are orthogonal.

I leave the proof of this being done with gaussian integrals as an exercise for the reviewer.

Exercise 7.3.4 Using Eqs. (7.3.23)-(7.3.25), show that

$$\begin{aligned} \langle n'|\, X\, |n\rangle &= \Big(\frac{\hbar}{2m\omega}\Big)^{1/2}[\delta_{n',n+1}(n+1)^{1/2} + \delta_{n',n-1}n^{1/2}] \\ \langle n'|\, P\, |n\rangle &= \Big(\frac{m\omega\hbar}{2}\Big)^{1/2} i[\delta_{n',n+1}(n+1)^{1/2} - \delta_{n',n-1}n^{1/2}] \end{aligned}$$

Consider the identities

$$H_n' = 2nH_{n-1} \tag{7.3.23}$$
$$2yH_n(y) = H_{n+1}(y) + 2nH_{n-1}(y) \tag{7.3.24}$$
$$\int_{-\infty}^{\infty} H_n(y)H_{n'}(y)e^{-y^2}dy = \delta_{n,n'}\sqrt{\pi}2^n n! \tag{7.3.25}$$

We begin by computing the first equation, starting from the LHS

$$\langle n'|\, X\, |n\rangle = \int \langle n'|x\rangle\, \langle x|\, X\, |n\rangle\, dx = \int \langle n'|x\rangle\, x\, \langle x|n\rangle\, dx = \int \psi_{n'}^*(x)x\psi_n(x)dx$$

The normalized solution of the harmonic oscillator are

$$\psi_n(x) = A_n \exp\Big(-\frac{(x\Delta)^2}{2}\Big)H_n(x\Delta)$$

Where A_n is a normalization constant equal to $(m\omega/\pi\hbar 2^{2n}(n!)^2)^{1/4}$ and Δ is a constant equal to $(m\omega/\hbar)^{1/2}$.

We proceed to compute the desired expectation value using $\psi_n(x)$. We will make use of the substitution $y = x\Delta$ and $dy = \Delta dx$ to make the use of identities (7.3.23) - (7.3.25) clear.

$$\begin{aligned}
\langle n'|\, X\, |n\rangle &= \frac{A_{n'}^* A_n}{\Delta^2}\int \exp\big(-y^2\big)H_{n'}^*(y)yH_n(y)dy \\
&= \frac{A_{n'}^* A_n}{2\Delta^2}\left(\int \exp(-y^2)H_{n'}^*(y)H_{n+1}(y)dy + \int \exp(-y^2)H_{n'}^*(y)2nH_{n-1}(y)dy\right) \\
&= \frac{A_{n'}^* A_n}{2\Delta^2}\left(\delta_{n',n+1}\sqrt{\pi}2^{n+1}(n+1)! + 2n\delta_{n',n-1}\sqrt{\pi}2^{n-1}(n-1)!\right)
\end{aligned}$$

Before proceeding, we note that within each term, we are allowed to interchange the indices of δ. We also realize that $(n+1)!/\sqrt{n!}\sqrt{(n+1)!} = \sqrt{n+1}$. We start by simplifying the common factor $A_{n'}^* * A_n/2\Delta^2$. We then arrive at the desired expression after substituting Δ back in to the equation.

$$\begin{aligned}
\langle n'|\, X\, |n\rangle &= \frac{1}{\Delta(2\pi)^{1/2}}\frac{1}{\sqrt{n'!n!2^{n'}2^{n+1}}}\left(\delta_{n',n+1}\sqrt{\pi}2^{n+1}(n+1)! + 2n\delta_{n',n-1}\sqrt{\pi}2^{n-1}(n-1)!\right) \\
&= \frac{1}{\Delta\sqrt{2}}\left(\frac{\delta_{n',n+1}2^{n+1}(n+1)!}{\sqrt{(n+1)!n!2^{n+1}2^{n+1}}} + \frac{\delta_{n',n-1}2^n n!}{\sqrt{(n-1)!n!2^{n-1}2^{n+1}}}\right) \\
&= \frac{1}{\Delta\sqrt{2}}\left(\delta_{n',n+1}(n+1)^{1/2} + \delta_{n',n-1}n^{1/2}\right)
\end{aligned}$$

The derivation for the matrix elements of P follow the same steps with a characteristic use of the $H_n'(y)$ identity to resolve the derivative generated by P.

$$\begin{aligned}
\langle n'|\, P\, |n\rangle &= \int \psi_{n'}^*(x)(-i\hbar)\psi_n'(x)dx \\
&= -i\hbar \int \big(A_{n'}^* e^{-y^2/2} H_{n'}(y)\big)\big(A_n H_n'(y)\Delta e^{-y^2/2} - A_n H_n(y) e^{-y^2/2}\Delta y\big)dx \\
&= -i\hbar A_{n'}^* A_n \left[\int e^{-y^2} H_{n'}(y) 2n H_{n-1}(y)dy - \int e^{-y^2} H_{n'}(y) y H_n(y)dy\right] \\
&= -i\hbar A_{n'}^* A_n \sqrt{\pi}\left[\delta_{n',n-1} 2^n n! - \frac{1}{2}\left(\delta_{n',n+1}2^{n+1}(n+1)! + 2n\delta_{n',n-1}2^{n-1}(n-1)!\right)\right] \\
&= i\hbar A_{n'}^* A_n \sqrt{\pi}\left[\delta_{n',n+1}2^n(n+1)! - \delta_{n',n-1}2^{n-1}n!\right] \\
&= \left(\frac{m\omega\hbar}{2}\right)^{1/2} i\left[\frac{\delta_{n',n+1}2^n(n+1)!\sqrt{2}}{\sqrt{2^{n+1}2^n(n+1)!n!}} - \frac{\delta_{n',n-1}2^{n-1}n!\sqrt{2}}{\sqrt{2^{n-1}2^n(n-1)!n!}}\right] \\
&= \left(\frac{m\omega\hbar}{2}\right)^{1/2} i[\delta_{n',n+1}(n+1)^{1/2} - \delta_{n',n-1}n^{1/2}]
\end{aligned}$$

Exercise 7.3.5 Using the symmetry arguments from Exercise 7.3.3 show that $\langle n|\, X\, |n\rangle = \langle n|\, P\, |n\rangle = 0$ and thus that $\langle X^2\rangle = (\Delta X)^2$ and $\langle P^2\rangle = (\Delta P)^2$ in these states. Show that $\langle 1|\, X^2\, |1\rangle = 3\hbar/2m\omega$ and $\langle 1|\, P^2\, |1\rangle = \frac{3}{2}m\omega\hbar$

We know that $\psi_n(x)$ are alternating even and odd functions for each n. Thus $\langle n|\, X\, |n\rangle = \int |\psi_n(x)|^2 x dx = 0$ because the functions $|\psi_n(x)|^2$ will always be even and x is always odd. Then by 7.3.3 this evaluates to zero. Similarly, $\langle n|\, P\, |n\rangle = (|A_n|^2/\Delta^2)\int \exp(-y^2)|H_n(y)|^2 y dy = 0$ because the functions $\exp(-y^2)|H_n(y)|^2$ is always positive and y is always odd, resulting in the zero result.

Since $(\Delta O)^2 = \langle O^2\rangle - \langle O\rangle^2$, the desired result of$\langle X^2\rangle = (\Delta X)^2$ and $\langle P^2\rangle = (\Delta P)^2$ immediately follows from $\langle X\rangle = 0$ and $\langle P\rangle = 0$ which was just shown.

To evaluate $\langle 1|\, X^2\, |1\rangle = 3\hbar/2m\omega$, we integrate the expectation value making use of $\psi_1(x)$ and gaussian integral expressions. For ease of computation, let $\alpha = m\omega/\hbar$

$$\begin{aligned}
\langle 1|\, x^2\, |1\rangle = \int |\psi_1(x)|^2 x^2 dx &= \frac{2\alpha^{3/2}}{\sqrt{\pi}}\int \exp(-\alpha x^2)x^4 dx \\
&= \frac{2\alpha^{3/2}}{\sqrt{\pi}}\frac{d^2}{d\alpha^2}\left(\int \exp(-\alpha x^2)dx\right) \\
&= \frac{2\alpha^{3/2}}{\sqrt{\pi}}\frac{d^2}{d\alpha^2}\left((\pi/\alpha)^{1/2}\right) \\
&= \frac{2\alpha^{3/2}}{\sqrt{\pi}}\left(\frac{3}{4}\frac{\sqrt{\pi}}{\alpha^{5/2}}\right) = \frac{3\hbar}{2m\omega}
\end{aligned}$$

Then for the momentum

$$\begin{aligned}
\langle 1|\, P^2\, |1\rangle &= -\hbar^2 \int \psi_1^*(x)\psi_1''(x)dx \\
&= -h^2 \frac{2\alpha^{3/2}}{\sqrt{\pi}} \int \Big(e^{-\alpha x^2/2}x\Big)\Big(e^{-\alpha x^2/2}(-3\alpha x + \alpha^2 x^3)\Big)dx \\
&= -h^2 \frac{2\alpha^{3/2}}{\sqrt{\pi}} \int e^{-\alpha x^2}(-3\alpha x^2 + \alpha^2 x^4)\Big)dx \\
&= -h^2 \frac{2\alpha^{3/2}}{\sqrt{\pi}} \left(\frac{-3\alpha}{2\alpha}\sqrt{\frac{\pi}{\alpha}} + \frac{3\alpha^2}{4}\sqrt{\frac{\pi}{\alpha^5}}\right) \\
&= -h^2 \frac{3\alpha}{2} = 3m\omega\hbar/2
\end{aligned}$$

Exercise 7.4.1 Compute the matrix elements of X and P in the $\langle n|$ basis and compare with the result from Exercise 7.3.4.

This question is implicitly asking us to use the creation and annihilation operators that was introduced during that section. The matrix element computation evaluates to the same expression, with noticeably fewer steps.

Making use of the identities $a\,|n\rangle = \sqrt{n}\,|n-1\rangle$ and $a^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$ and also using the expression for X and P as equations of a and $a^\dagger$ (see 7.4.28 & 7.4.29), we evaluate,

$$\begin{aligned}
\langle n'|\, X\, |n\rangle &= \langle n'|\, \sqrt{\hbar/2m\omega}\big[a + a^\dagger\big]\, |n\rangle \\
&= \sqrt{h/2m\omega}\big(\, \langle n'|\, a\, |n\rangle + \langle n'|\, a^\dagger\, |n\rangle\, \big) \\
&= \Big(\frac{\hbar}{2m\omega}\Big)^{1/2}[\delta_{n',n+1}(n+1)^{1/2} + \delta_{n',n-1}n^{1/2}] \\
\langle n'|\, P\, |n\rangle &= \langle n'|\, i\sqrt{m\omega\hbar/2}\big[a^\dagger - a\big]\, |n\rangle \\
&= i\sqrt{m\omega\hbar/2}\big(\, \langle n'|\, a^\dagger\, |n\rangle - \langle n'|\, a\, |n\rangle\, \big) \\
&= \Big(\frac{m\omega\hbar}{2}\Big)^{1/2} i[\delta_{n',n+1}(n+1)^{1/2} - \delta_{n',n-1}n^{1/2}]
\end{aligned}$$

Exercise 7.4.2 Find $\langle X\rangle$, $\langle P\rangle$, $\langle X^2\rangle$, $\langle P^2\rangle$, $\Delta X \cdot \Delta Y$, in the state $|n\rangle$.

$\langle X\rangle$, $\langle P\rangle$ are trivially zero using the expression from 7.4.1. As the dirac deltas will have indices $\delta_{n,n+1}$ and $\delta_{n,n-1}$ which are never non-zero.

Using the same expression, $\langle X^2\rangle = \frac{\hbar}{2m\omega}\,\langle n|\,(a^2 + (a^\dagger)^2 + aa^\dagger + a^\dagger a)\,|n\rangle$. Since we only care about indices that have no net change on $|n\rangle$, only the last two terms are nonzero. then $\langle X^2\rangle = \frac{\hbar}{2m\omega}\big(2n{+}1\big)$. Similarly $\langle P^2\rangle = -(m\omega\hbar/2)\,\langle n|\,((a^\dagger)^2{+}a^2{-}a^\dagger a{-}aa^\dagger)\,|n\rangle = \frac{m\omega\hbar}{2}(2n{+}1)$.

Thus,

$$\Delta X \cdot \Delta Y = \sqrt{\langle X^2\rangle - \langle X\rangle^2}\sqrt{\langle Y^2\rangle - \langle Y\rangle^2} = \frac{\hbar}{2}(2n+1)$$

Exercise 7.4.3 (Virial theorem). The virial theorem in classical mechanics states that for a particle bound by a potential $V(r) = ar^k$, the average (over the orbit) kinetic and potential energies are related by

$$\overline{T} = c(k)\overline{V}$$

when $c(k)$ only depends on k. Show that $c(k) = k/2$ by considering a circular orbit. Using the results from the previous exercise show that for the oscillator ($k = 2$)

$$\langle T\rangle = \langle V\rangle$$

in the quantum states $|n\rangle$]

Circular motion also requires $mr\dot{\theta}^2 = dV/dr = akr^{k-1}$ so thus $\dot{\theta}^2 = akr^{k-2}/m$. We also note that in circular coordinates $\overline{T} = (1/2)mr^2\dot{\theta}^2$. Note that r and $\dot{\theta}$ are constant in time for uniform circular motion. We proceed to evaluate both averages using an integral over time from 0 to τ.

$$\begin{aligned}\int_0^\tau \frac{1}{2}mr^2\dot{\theta}^2 dt &= c(k)\int_0^\tau ar^k dt \\ \frac{1}{2}mr^2\dot{\theta}^2\tau &= c(k)ar^k\tau \\ \frac{1}{2}r^2akr^{k-2} &= c(k)ar^k \\ c(k) &= k/2\end{aligned}$$

.

For the quantum harmonic oscillator we can easily equate the desired expression by using the calculations for $\langle X^2\rangle$ and $\langle P^2\rangle$.

$$\begin{aligned}\langle T\rangle &= \langle V\rangle \\ \langle P^2/2m\rangle &= \langle m\omega^2X^2/2\rangle \\ (1/2m)(m\omega\hbar/2)(2n+1) &= (m\omega^2/2)(\hbar/2m\omega)(2n+1) \\ (\hbar\omega/4)(2n+1) &= (\hbar\omega/4)(2n+1)\end{aligned}$$

Exercise 7.4.4 Show that $\langle n|\, X^4\, |n\rangle = (\hbar/2m\omega)^2[3+6n(n+1)]$.

$$\begin{aligned}\langle n|\, X^4\, |n\rangle &= \langle n|\, (\hbar/2m\omega)^2(a^\dagger + a)^4\, |n\rangle \\ &= (\hbar/2m\omega)^2\, \langle n|\, (a^\dagger)^2a^2 + a^2(a^\dagger)^2 + a^\dagger aa^\dagger a + aa^\dagger aa^\dagger + a^\dagger a^2a^\dagger + a(a^\dagger)^2a\, |n\rangle \\ &= (\hbar/2m\omega)^2([n^2-n] + [n^2+3n+2] + [n^2] + [n^2+2n+1] + [n^2+1] + [n^2+1]) \\ &= (\hbar/2m\omega)^2(3+6n(n+1))\end{aligned}$$

Exercise 7.4.5 At $t=0$ a particle starts out in $|\psi(0)\rangle = 1/2^{1/2}(|0\rangle + |1\rangle)$. (1) Find $|\psi(t)\rangle$; (2) find $\langle X(0)\rangle = \langle\psi(0)|\, X\, |\psi(0)\rangle\, , \langle P(0)\rangle\, , \langle X(t)\rangle\, , \langle P(t)\rangle$; (3) find $\langle\dot{X}(t)\rangle$ and $\langle\dot{P}(t)\rangle$ using Ehrenfest's theorem and solve for $\langle X(t)\rangle$ and $\langle P(t)\rangle$ and compare with part (2).

We can solve for $|\psi(t)\rangle$ by multiplying the initial wave function by the propagator.

$$\begin{aligned}|\psi(t)\rangle = U(t)\, |\psi(0)\rangle &= \frac{1}{\sqrt{2}}\exp(-iHt/\hbar)(|0\rangle + |1\rangle) \\ &= \frac{1}{\sqrt{2}}\left(e^{-iE_0t/\hbar}\, |0\rangle + e^{-iE_1t/\hbar}\, |1\rangle\right) \\ &= \frac{1}{\sqrt{2}}e^{-iE_0t/\hbar}\left(|0\rangle + e^{-i\omega t}\, |1\rangle\right)\end{aligned}$$

The next steps are computed as such, where the global phases consistently cancel out. I also am omitting kets that are larger than $n = 1$ as they don't contribute to anything in the inner product.

$$\begin{aligned}
\langle X(0)\rangle &= \frac{1}{2}(\langle 0| + \langle 1|)\sqrt{\frac{\hbar}{2m\omega}}(a^\dagger + a)(|0\rangle + |1\rangle \\
&= \frac{1}{2}\sqrt{\frac{\hbar}{2m\omega}}(\langle 0| + \langle 1|)(|0\rangle + |1\rangle) \\
&= \sqrt{\frac{\hbar}{2m\omega}} \\
\langle X(t)\rangle &= \frac{1}{2}\big(\langle 0| + e^{i\omega t}\langle 1|\big)\left(\sqrt{\frac{\hbar}{2m\omega}}(a^\dagger + a)\right)\big(|0\rangle + e^{-i\omega t}|1\rangle\big)\big) \\
&= \frac{1}{2}\sqrt{\frac{\hbar}{2m\omega}}\big(\langle 0| + e^{i\omega t}\langle 1|\big)\big(|1\rangle + e^{-i\omega t}|0\rangle\big)\big) \\
&= \sqrt{\frac{\hbar}{2m\omega}}\cos(\omega t) \\
\langle P(0)\rangle &= \frac{1}{2}(\langle 0| + \langle 1|)i\sqrt{\frac{\hbar m\omega}{2}}(a^\dagger - a)(|0\rangle + |1\rangle \\
&= i\frac{1}{2}\sqrt{\frac{\hbar m\omega}{2}}(\langle 0| + \langle 1|)(|1\rangle - |0\rangle) \\
&= 0 \\
\langle P(t)\rangle &= \frac{1}{2}\big(\langle 0| + e^{i\omega t}\langle 1|\big)\left(i\sqrt{\frac{\hbar m\omega}{2}}(a^\dagger - a)\right)\big(|0\rangle + e^{-i\omega t}|1\rangle\big)\big) \\
&= \frac{1}{2}i\sqrt{\frac{\hbar m\omega}{2}}\big(\langle 0| + e^{i\omega t}\langle 1|\big)\big(|1\rangle - e^{-i\omega t}|0\rangle\big)\big) \\
&= -\sqrt{\frac{\hbar m\omega}{2}}\sin(\omega t)
\end{aligned}$$

For the next part we can arrive at the same solution by using Ehrenfest's theorem and solving the differential equation.

$$\begin{aligned}
\langle \dot{X}(t)\rangle &= \langle P(t)\rangle / m \\
\langle \dot{P}(t)\rangle &= -m\omega^2\langle X\rangle
\end{aligned}$$

These two statements can be combined to become a second order differential equation, solved by typical sines/cosines. We can use the initial conditions from part 2 to evaluate to the same $\langle X(t)\rangle$ and $\langle P(t)\rangle$ expression

$$\begin{aligned}
\langle \ddot{X}(t)\rangle &= -\omega^2 \langle X\rangle \\
\langle X(t)\rangle &= A\cos(\omega t) + B\sin(\omega t) \\
\langle X(t)\rangle &= \sqrt{\hbar/2m\omega}\cos(\omega t) \\
\langle \ddot{P}(t)\rangle &= -w^2 \langle P\rangle \\
\langle P(t)\rangle &= -\sqrt{\hbar m\omega/2}\sin(wt)
\end{aligned}$$

Exercise 7.4.6 Show that $\langle a(t)\rangle = \exp(-i\omega t)\langle a(0)\rangle$ and that $\langle a^\dagger(t)\rangle = \exp(i\omega t)\langle a^\dagger(0)\rangle$

the creation (as well as it's adjoint) can we expressed in terms of X and P, hence this question asking about how they can be time dependent. We can use the answers to exercise 7.4.5 in order to evaluate both sides of the equation and see their equivalence.

$$\begin{aligned}
\langle a(t)\rangle &= \exp(-i\omega t)\langle a(0)\rangle \\
\big(\frac{m\omega}{2\hbar}\big)^{1/2}\langle X(t)\rangle + i\big(\frac{1}{2m\omega\hbar}\big)^{1/2}\langle P(t)\rangle &= \exp(-i\omega t)\left[\big(\frac{m\omega}{2\hbar}\big)^{1/2}\langle X(0)\rangle + i\big(\frac{1}{2m\omega\hbar}\big)^{1/2}\langle P(0)\rangle\right] \\
\frac{1}{2}\cos(\omega t) - \frac{1}{2}i\sin(\omega t) &= \frac{1}{2}\exp(-i\omega t)
\end{aligned}$$

The adjoint version of the expression follows directly from taking the adjoint of the above.

Exercise 7.4.7 Verify Eq. (7.4.40) for the case

$$\begin{cases} \Omega = X & ,\Lambda = X^2 + P^2 \\ \Omega = X^2 & ,\Lambda = P^2 \end{cases}$$

We need to verify the following expression

$$[\Omega(X,P),\Lambda(X,P)] = i\hbar\Gamma(X,P) \tag{7.4.40}$$

We make repeated use of commutator identities

$$\begin{aligned}
[X, X^2+P^2] &= [X,P^2] \\
&= P[X,P] + [X,P]P \\
&= i\hbar(2P) \\
[X^2,P^2] &= P[X^2,P] + [X^2,P]P \\
&= P\big(X[X,P] + [X,P]X\big) + \big(X[X,P] + [X,P]X\big)P \\
&= P(i\hbar 2X) + (i\hbar 2X)P \\
&= i\hbar(2(XP+PX))
\end{aligned}$$

Exercise 7.4.8 Consider the three angular momentum variables in classical mechanics:

$$\begin{aligned}
l_x &= yp_z - zp_y \\
l_y &= zp_x - xp_z \\
l_z &= xp_y - yp_x
\end{aligned}$$

(1) Construct L_x, L_y, and L_z, the quantum counterparts, and note that there are no ordering ambiguities.

The quantum operators of these dependent variables can be constructed with $x \to X$ (and for all other spatial directions), and $p \to P$.

$$\begin{aligned} L_x &= YP_z - ZP_y \\ L_y &= ZP_x - XP_z \\ L_z &= XP_y - YP_x \end{aligned}$$

(2) Verify that $\{l_x, l_y\} = l_z$ [See equation (2.7.3) for definition of Poisson brackets].

The sum can be greatly simplified by noticing that $\partial l_i/\partial q_i = \partial l_i/\partial p_i = 0$

$$\begin{aligned} \{l_x(y,z,p_y,p_z), l_y(x,z,p_x,p_z)\} &= \sum_i \left(\frac{\partial l_x}{\partial q_i}\frac{\partial l_y}{\partial p_i} - \frac{\partial l_x}{\partial p_i}\frac{\partial l_y}{\partial q_i} \right) \\ &= \frac{\partial l_x}{\partial z}\frac{\partial l_y}{\partial p_z} - \frac{\partial l_x}{\partial p_z}\frac{\partial l_y}{\partial z} \\ &= xp_y - yp_x = l_z \end{aligned}$$

(3) Verify that $[L_x, L_y] = i\hbar L_z$.

The expression can be expanded and then distribute to produce 8 terms. The trick to neatly evaluating all these expressions is by repeatedly using the identity $[AB, CD] = [A, C]BD + C[A, D]B + A[B, C]D + CA[B, D]$ and note from (7.4.39) that the only nonzero commutator will result from when $[X_i, P_i] = i\hbar$

$$\begin{aligned} L_xL_y - L_yL_x &= (YP_z - ZP_y)(ZP_x - XP_z) - (ZP_x - XP_z)(YP_z - ZP_y) \\ &= \quad YP_zZP_x - YP_zXP_z - ZP_yZP_x + ZP_yXP_z \\ &\quad -ZP_xYP_z + ZP_xZP_y + XP_zYP_z - XP_zZP_y \\ &= [YP_z, ZP_x] + [ZP_y, XP_z] + [XP_z, YP_z] + [ZP_x, ZP_y] \\ &= Y[P_z, Z]P_x + X[Z, P_z]P_y \\ &= i\hbar(XP_y - YP_x) = i\hbar L_z \end{aligned}$$

Exercise 7.4.9 Consider the unconventional (but fully acceptable) operator choice

$$\begin{aligned} X &\to x \\ P &\to -i\hbar\frac{d}{dx} + f(x) \end{aligned}$$

in the X basis.

(1) Verify that the canonical commutation relation is satisfied.

$$[X, P] = [x, -i\hbar\frac{d}{dx} + f(x)] = [x, -i\hbar\frac{d}{dx}] + [x, f(x)] = i\hbar$$

(2) It is possible to interpret the change int he oeprator assignemnt as a result of a unitary change of the X basis:

$$|x\rangle \to |\tilde{x}\rangle = e^{ig(X)/\hbar}\,|x\rangle = e^{ig(x)/\hbar}\,|x\rangle$$

where $g(x) = \int^x f(x')dx'$. We first verify that $\langle\tilde{x}|\,X\,|\tilde{x}'\rangle = x\delta(x-x')$. Using this given definition of $|\tilde{x}\rangle$, we can substitute it into the LHS of the expression to start.

$$\begin{aligned}\langle\tilde{x}|\,X\,|\tilde{x}'\rangle &= e^{-ig(x)/\hbar}\,\langle x|\,X\,|x'\rangle\,e^{ig(x')/\hbar}\\ &= e^{-ig(x)/\hbar}x\delta(x-x')e^{ig(x')/\hbar}\\ &= x\delta(x-x')\end{aligned}$$

The last step comes from the delta function enforcing $x' = x$.

Next we verify the momentum matrix elements in the new basis

$$\begin{aligned}\langle\tilde{x}|\,P\,|\tilde{x}'\rangle &= e^{-ig(x)/\hbar}\,\langle x| - i\hbar\frac{d}{dx} + f(x)\,|x'\rangle\,e^{ig(x')/\hbar}\\ &= e^{-ig(x)/\hbar}\bigg[-i\hbar\frac{d}{dx}\delta(x-x') + f(x)\delta(x-x')\bigg]e^{ig(x')/\hbar}\\ &= e^{-ig(x)/\hbar}\bigg[-i\hbar\frac{d}{dx}\bigg(\delta(x-x')e^{ig(x')/\hbar}\bigg) + f(x)\delta(x-x')e^{ig(x')/\hbar)}\bigg]\\ &= -i\hbar\bigg[\frac{d}{dx}\delta(x-x') + f(x)\delta(x-x')\bigg]\\ &= \bigg[-i\hbar\frac{d}{dx} + f(x)\bigg]\delta(x-x')\end{aligned}$$

Where the second to last line is arrived at from the exponentials canceling the right most exponential is distributed to the terms and the delta functions enforce $x = x'$, this is done with particular ease because the derivative is with respect to only x and not x', so we can let the exponentials pass the derivatives, conjoin together as an overall phase that cancels to 1 after we enforce $x = x'$

Exercise 7.4.10 This question has a lot of context which will not be posted here. The general context is considering a particle in two dimensions and it's hamiltonian. The first thing to illustrate is that if you wanted to express a Hamiltonian in non-Cartesian coordinates, one cannot start with a classical Hamiltonian expressed in non-Cartesian coordinates and then promote classical variables to quantum operators. It is, however, possible to start with a classical Hamiltonian written in Cartesian coordinates, promote the classical variables to quantum operators, and then convert that Hamiltonian to the preferred non-Cartesian coordinates.

(1) One quick check that the question asks to verify is the expression for the classical Hamiltonian of a two dimensional particle written in polar coordinates. The classical expression is

$$\mathcal{H} = \frac{p_x^2 + p_y^2}{2m} + a(x^2+y^2)^{1/2}$$

We can arrive at the desired polar-coordinate $\mathcal{H}$ by multiplying ρ^2/ρ^2 and adding a net-zero cross term $2xyp_xp_y$.

$$\begin{aligned}\mathcal{H} &= \frac{p_x^2\rho^2 + p_y^2\rho^2 + 2xyp_xp_y\rho^2 - 2xyp_xp_y\rho^2}{2m\rho^2} + a\rho \\ &= \frac{(xp_x + yp_y)^2 + (xp_y - yp_x)^2}{2m\rho^2} + a\rho = \frac{p_\rho^2}{2m} + \frac{p_\phi^2}{2m\rho^2} + a\rho\end{aligned}$$

(2) The next part of the problem observes how promoting this Hamiltonian via $P_\rho \to -i\hbar\frac{\partial}{\partial\rho}$ leads to an incorrect quantum Hamiltonian, noticeably because it is not a Hermitian operator. We now want to verify that the choosing $P_\rho \to -i\hbar\left(\frac{\partial}{\partial\rho} + \frac{1}{2\rho}\right)$ does satisfy the canonical commutation rule and it is Hermitian.

First, the canonical commutation rule states that a coordinate operator and it's conjugate momentum operator commute to $i\hbar$. In these polar coordinates, we can use ρ and P_ρ to demonstrate the relation. As usual, to compute the commutator, we utilize a test function $f(\rho)$.

$$\begin{aligned}[\rho, P_\rho]f(\rho) &= \left[\rho\left(-i\hbar(\frac{\partial}{\partial\rho} + \frac{1}{2\rho})\right) - \left(-i\hbar(\frac{\partial}{\partial\rho} + \frac{1}{2\rho})\right)\rho\right]f(\rho) \\ &= \left[-i\hbar\rho\frac{\partial f(\rho)}{\partial\rho} - \frac{i\hbar f(\rho)}{2} + i\hbar\frac{\partial(\rho f(\rho))}{\partial\rho} + \frac{i\hbar f(\rho)}{2}\right] \\ &= i\hbar\left(-\rho\frac{\partial f(\rho)}{\partial\rho} + +p\frac{\partial f(\rho)}{\partial\rho} + f(\rho)\right) = (i\hbar)f(\rho)\end{aligned}$$

The next step is to check whether the operator is Hermitian. In other words, is $\langle P_\rho\psi_1|\psi_2\rangle = \langle\psi_1|P_\rho\psi_2\rangle$. Starting from the left hand side, we evaluate the inner product in polar coordinates.

$$\begin{aligned}\langle P_\rho\psi_1|\psi_2\rangle &= \int_0^{2\pi} d\phi\int_0^\infty \rho d\rho\left[\left(-i\hbar(\frac{\partial}{\partial\rho} + \frac{1}{2\rho})\psi_1\right)^*\psi_2\right] \\ &= i\hbar\int_0^{2\pi} d\phi\int_0^\infty \rho d\rho\left[\frac{\partial\psi_1^*}{\partial\rho} + \frac{\psi_1^*}{2\rho}\right]\psi_2\end{aligned}$$

The first term inside the integral can be evaluated by parts with $u = \rho\psi_2$ and $dv = d\rho(\partial\psi_1^*/\partial\rho)$. We can also make use of the assumption that $\rho\psi_1^*\psi_2 \to$ as $\rho \to 0$ or ∞ in order to cancel one of the terms. The expression then evaluates to become the hermitian

conjugate,

$$
\begin{aligned}
&= i\hbar \int_0^{2\pi} d\phi \left[\left[\rho \psi_1^* \psi_2\right]_0^\infty - \int_0^\infty d\rho \psi_1^* \big(\psi_2 + \rho \frac{\partial \psi_2}{\partial \rho}\big) \right] + i\hbar \int_0^{2\pi} d\phi \int_0^\infty d\rho \frac{\psi_1^* \psi_2}{2} \\
&= -i\hbar \int_0^{2\pi} d\phi \int_0^\infty d\rho \left[\psi_1^* \psi_2 + \psi_1^* \rho \frac{\partial \psi_2}{\partial \rho} - \frac{1}{2} \psi_1^* \psi_2 \right] \\
&= -i\hbar \int_0^{2\pi} d\phi \int_0^\infty d\rho \left[\psi_1^* \rho \frac{\partial \psi_2}{\partial \rho} + \frac{1}{2} \psi_1^* \psi_2 \right] \\
&= \int_0^{2\pi} d\phi \int_0^\infty \rho d\rho \left[\psi_1^* \bigg(-i\hbar \big(\frac{\partial}{\partial \rho} + \frac{1}{2\rho}\big) \psi_2 \bigg) \right] = \langle \psi_1 | P_\rho \psi_2 \rangle
\end{aligned}
$$

(3) In the cartesian case we saw that adding an $f(x)$ to $-i\hbar\partial/\partial x$ didn't have any physical effect, whereas here the addition of a function of ρ to $-i\hbar\partial/\partial\rho$ seems important. Why?

We notice that in order for $P \to -i\hbar\frac{d}{dx} + f(x)$ to be hermitian, we require f(x) to be real. This is because $P^\dagger = -i\hbar\frac{d}{dx} + f^\dagger(x)$ and $P^\dagger = P$ if and only if $f^\dagger(x) = f(x)$, (f is real).

In the case where $P_\rho \to -i\hbar\frac{\partial}{\partial\rho} - \frac{i\hbar}{2\rho}$, the additional function is purely imaginary. In fact, it is specifically chosen because in evaluating this P's hermiticity, we have to evaluate it's inner product in polar coordinates whose integration elements includes an extra ρ. This ρ is consequential in so far that it produces an extra term when the derivative wrt ρ acts on it multiplied against a wave function. This extra term can only be canceled out to neatly make P hermitian if the additinal function is of this exact form. This is demonstrated in part (2). Thus, in this case, there is only one function of ρ that could be added to $-i\hbar\partial_\rho$ to make P hermitian.

(4) Feed in the new momentum operator P_ρ and show that

$$
H \to \frac{-\hbar^2}{2m} \left(\frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial}{\partial \rho} - \frac{1}{4\rho^2} + \frac{1}{\rho^2}\frac{\partial^2}{\partial \phi^2} \right) + a\rho
$$

This next step is a simply substitution and putting H on a test function to evaluate the derivatives.

$$
\begin{aligned}
Hf &= \left[\frac{P_\rho^2}{2m} + \frac{P_\phi^2}{2m\rho^2} + a\rho \right] f = \left[\frac{\big(-i\hbar \left(\frac{\partial}{\partial\rho} + \frac{1}{2\rho} \right) \big)^2}{2m} + \frac{\big(-i\hbar \frac{\partial}{\partial\rho} \big)^2}{2m\rho^2} + a\rho \right] f \\
&= \frac{-\hbar^2}{2m} \left[\frac{\partial^2}{\partial\rho^2} + \frac{1}{\rho^2}\frac{\partial^2}{\partial\phi^2} \right] f + \frac{-\hbar^2}{2m} \left[\frac{f}{4\rho^2} + \frac{\partial(f/2\rho)}{\partial\rho} + \frac{1}{2\rho}\frac{\partial f}{\partial\rho} \right] + a\rho f \\
&= \frac{-\hbar^2}{2m} \left[\frac{\partial^2}{\partial\rho^2} + \frac{1}{\rho^2}\frac{\partial^2}{\partial\phi^2} + a\rho \right] f + \frac{-\hbar^2}{2m} \left[\frac{1}{\rho}\frac{\partial}{\partial\rho} - \frac{1}{4\rho^2} \right] f + a\rho f
\end{aligned}
$$

Moving terms around, we arrive at the desired Hamiltonian.

Exercise 7.5.1 Project $a\,|0\rangle = 0$ on to the P basis and obtain $\psi_0(p)$.

In the momentum basis,

$$X \to i\hbar \frac{d}{dp}$$
$$P \to p$$

We can represent the annihilation operator in that basis and project $a|0\rangle = 0$ on to the p basis. It is easiest to compute by substituting $\alpha = (m\omega\hbar)^{1/2}$ within the annihilation operator.

$$\begin{aligned} 0 = \langle p| a |0\rangle &= \langle p| \left(\frac{i}{\sqrt{2}} \left[\alpha \frac{d}{dp} + \frac{p}{\alpha} \right] \right) |0\rangle \\ &= \alpha \left\langle p \middle| \frac{d}{dp} 0 \right\rangle + \frac{p}{\alpha} \left\langle p \middle| 0 \right\rangle \\ &= \alpha \psi_0'(p) + \frac{p\psi_0(p)}{\alpha} \end{aligned}$$

This is a familiar differential equation that has general solution

$$\psi_0 = A_0 \exp(-p^2/2\alpha^2)$$

We evaluate $\langle \psi_0 | \psi_0 \rangle$ using a standard gaussian integral to find the value of A_0.

$$1 = \langle \psi_0 | \psi_0 \rangle = \int_{-\infty}^{\infty} dp |A_0|^2 \exp(-p^2/\alpha^2) = |A_0|^2 (\pi\alpha^2)^{1/2}$$

Solving for A_0, we can express the wave function as

$$\psi_0(p) = (m\omega\hbar\pi)^{-1/4} \exp(-p^2/2m\omega\hbar)$$

Exercise 7.5.2 Project the relation

$$a|n\rangle = n^{1/2}|n-1\rangle$$

on the X basis and derive the recursion relation

$$H_n'(y) = 2nH_{n-1}(y)$$

When we project, it is useful to write down $\psi_n(x)$'s full expression. For the ease of canceling terms out later, we let $\alpha = \sqrt{m\omega/\hbar}$, and $f(n) = 2^{2n}(n!)^2$.

$$\psi_n(x) = \left(\frac{\alpha^2}{\pi f(n)} \right)^{1/4} \exp(-\alpha^2 x^2/2) H_n(\alpha x)$$

The derivative can also be written, using chain rule, as

$$\psi_n'(x) = \left(\frac{\alpha^2}{\pi f(n)} \right)^{1/4} \exp(-\alpha^2 x^2/2) \left[-\alpha^2 x H_n(\alpha x) + \alpha H_n'(\alpha x) \right]$$

So thus we project the relation on to the x basis using a in the x basis.

$$\langle x| \frac{1}{\sqrt{2}}\left(\alpha x + \frac{1}{\alpha}\frac{d}{dx}\right)|n\rangle = n^{1/2}\psi_{n-1}(x)$$
$$\frac{1}{\sqrt{2}}\left(\alpha x\psi_n(x) + \frac{1}{\alpha}\psi_n'(x)\right) = n^{1/2}\psi_{n-1}(x)$$

When we plug in our expressions for $\psi_n(x)$, $\psi_{n-1}(x)$, and $\psi_n'(x)$. All the terms share the factors $(\frac{\alpha^2}{\pi})^{1/4}\exp(-\alpha^2x^2/2)$ so they can be canceled out to leave the expression,

$$\frac{1}{\sqrt{2}}\left(\frac{\alpha x}{f(n)^{1/4}}H_n(\alpha x) + \frac{1}{\alpha f(n)^{1/4}}\Big[-\alpha^2 x H_n(\alpha x) + \alpha H_n'(\alpha x)\Big]\right) = n^{1/2}\frac{1}{f(n-1)^{1/4}}H_{n-1}(\alpha x)$$

The $H_n(\alpha x)$ terms cancel and we can multiply everything by $f(n)^{1/4}$ to simplify. Note that $(f(n)/f(n-1))^{1/4} = n\sqrt{2}$. If we call $y = \alpha x$ we arrive at the desired recursion relation.

$$\frac{1}{\sqrt{2}}H_n'(\alpha x) = n\sqrt{2}H_{n-1}(\alpha x)$$
$$H_n'(y) = 2nH_{n-1}(y)$$

Exercise 7.5.3 Starting with

$$a^\dagger + a = 2^{1/2}y$$

and

$$(a + a^\dagger)|n\rangle = n^{1/2}|n-1\rangle + (n+1)^{1/2}|n+1\rangle$$

and the normalized solutions of $\psi_n(x)$, derive the relation

$$H_{n+1}(y) = 2yH_n(y) - 2nH_{n-1}(y)$$

This next question follows a very similar strategy as the previous question. We first substitute the first equation in to the second and then project it on to the x basis.

$$y\sqrt{2}\psi_n(x) = \sqrt{n}\psi_{n-1}(x) + \sqrt{n+1}\psi_{n+1}(x)$$

Then the same factors cancel to only leave the $f(n)$ terms and the Hermite polynomials. We also want to multiply both sides so that we can get $H_{n+1}(y)$ with a unit factor (like the recursion relation does). Note that $(f(n+1)/f(n))^{1/4} = \sqrt{2}\sqrt{n+1}$ and $(f(n+1)/f(n-1))^{1/4} = 2\sqrt{n+1}\sqrt{n}$.

$$y\sqrt{2}\frac{H_n(y)}{f(n)^{1/4}} = \sqrt{n}\frac{H_{n-1}(y)}{f(n-1)^{1/4}} + \sqrt{n+1}\frac{H_{n+1}(y)}{f(n+1)^{1/4}}$$
$$\sqrt{n+1}H_{n+1}(y) = 2y\sqrt{n+1}H_n(y) - 2n\sqrt{n+1}H_{n-1}(y)$$
$$H_{n+1}(y) = 2yH_n(y) - 2nH_{n-1}(y)$$

Exercise 7.5.4 Thermodynamics of Oscillators. The Boltzman formula

$$P(i) = e^{-\beta E(i)}/Z$$

where

$$Z = \sum_i e^{-\beta E(i)}$$

gives the probability of finding a system in a state i with energy $E(i)$, when it is in thermal equilibrium with a reservoir of absolute temperature $T = 1/\beta k$, $k = 1.4 \times 10^{-16}$ ergs/degree K; being Boltzman's constant. (The "probability" referred to above is in relation to a classical ensemble of similar systems and has nothing to do with quantum mechanics.)

(1) Show that the thermal average of the system's energy is

$$\overline{E} = \sum_i E(i)P(i) = -\frac{\partial}{\partial \beta} \ln Z$$

This is accomplished by starting from the definition of a thermal average (weighted sum) and then recognizing the form of an exponential derivative and a logarithmic derivative (with chain rule).

$$\begin{aligned}
\overline{E} &= \sum_i E(i)P(i) = \frac{1}{Z}\sum_i E(i)e^{-\beta E(i)} \\
&= -\frac{1}{Z}\sum_i \frac{\partial}{\partial \beta} e^{-\beta E(i)} = -\frac{1}{Z}\frac{\partial}{\partial \beta}\sum_i e^{-\beta E(i)} \\
&= -\frac{1}{Z}\frac{\partial}{\partial \beta} Z = -\frac{\partial}{\partial \beta}\ln Z
\end{aligned}$$

(2) Let the system be a classical oscillator. The index i is now continuous and corresponds to the variables x and p describing the state of the oscillator, i.e.,

Arrive at Z_{cl} by changing from index i to x, p. This turns $\sum_i \to \int\int dxdp$ and $E(i) \to E(x,p) = p^2/2m + m\omega^2 x^2/2$.

All we have to do is make the necessary substitutions into the definition of Z.

$$\begin{aligned}
Z = \sum_i e^{-\beta E(i)} &\to \int\int dxdp e^{-\beta(p^2/2m + m\omega^2 x^2/2} \\
&= \int dx e^{-\beta m\omega^2 x^2/2} \int dp e^{-\beta p^2/2m} \\
&= (\pi/(\beta m\omega^2/2))^{1/2}(\pi/(\beta/2m))^{1/2} = 2\pi/\omega\beta
\end{aligned}$$

To find the classical expression for energy E_{cl} we just follow the same calculations for the thermal average of the system's energy.

$$\overline{E_{cl}} = -\frac{\partial}{\partial \beta}\ln(2\pi/\omega\beta) = -\frac{\omega\beta}{2\pi}\frac{2\pi}{\omega}(-\beta^{-2}) = 1/\beta = kT$$

(3) For the quantum oscillator the quantum number n plays the role of the index i. Show Z_{qu} and $\overline{E_{cl}}$.

We start this calculation by recalling the energy of a single quantum harmonic oscillator $E_n = (n + 1/2)\hbar\omega$, and substituting it in to the Z definition.

$$Z = \sum_i e^{-\beta E(i)} \to Z_{cl} = \sum_n e^{-\beta E(n)} = \sum_n e^{-\beta(n+1/2)\hbar\omega}$$
$$= e^{-\beta\hbar\omega/2}\left(\sum_n e^{-\beta\hbar\omega n}\right)$$

THe last term is simplified into the final expression by noticing the term inside the parenthesis is the infinite sum of a geometric series with initial value 1 and geometric ratio $r = e^{-\beta\hbar\omega}$. Thus

$$Z_{qu} = e^{-\beta\hbar\omega/2}(1 - e^{-\beta\hbar\omega})^{-1}$$

Then similarly E_{qu} is calculated through the same derivative expression

$$\overline{E_{qu}} = -\frac{\partial}{\partial\beta}\ln\left(\frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}\right) = -\frac{\partial}{\partial\beta}\Big[-\beta\hbar\omega/2 - \ln(1 - e^{-\beta\hbar\omega})\Big]$$
$$= \hbar\omega/2 + (1 - e^{-\beta\omega\hbar})^{-1}(\hbar\omega e^{-\beta\omega\hbar}) = \hbar\omega\left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1}\right)$$

(4) It is intuitively clear that as the temperature T increases (and $\beta = 1/kT$ decreases) the oscillator will get more and more excited and eventually (from the correspondence principle)

$$\overline{E_{cl}} \to_{T\to\infty} \to \overline{E_{qu}}$$

Verify that this is indeed true and show that "large T" means $T \gg \hbar\omega/k$.

This limit is not straightforward since both $\overline{E_{qu}}$ and $\overline{E_{cl}}$ diverge as $\beta \to 0$. Our hint is that the classical expression is linear in β, so we can expand the quantum analogue and only keep terms of linear order or lower.

$$\overline{E_{qu}} = \hbar\omega\left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1}\right)$$
$$= \hbar\omega\left(\frac{1}{2} + \frac{1}{(\beta\hbar\omega) + \frac{1}{2}(\beta\hbar\omega)^2 + O(\beta\hbar\omega)^3}\right)$$
$$= \hbar\omega\left(\frac{1}{2} + \frac{1}{\beta\hbar\omega}\frac{1}{1 + \frac{1}{2}(\beta\hbar\omega) + O(\beta\hbar\omega)^2}\right)$$

At this point we can make use of the expansion $\frac{1}{1+y} = 1 - y + y^2 - y^3...$ and we let $y = \frac{1}{2}(\beta\hbar\omega) + O(\beta\hbar\omega)^2$. If we keep only terms constant and linear in β, we further simplify the expression.

$$= \hbar\omega\left(\frac{1}{2} + \frac{1}{\beta\hbar\omega}\left[1 - \left(\frac{1}{2}(\beta\hbar\omega) + O(\beta\hbar\omega)^2\right) + O(y^2)\right]\right)$$
$$= \hbar\omega\left(\frac{1}{2} + \frac{1}{\beta\hbar\omega}\left[1 - \frac{1}{2}\beta\hbar\omega\right]\right) = \frac{1}{\beta} = \overline{E_{cl}}$$

The evaluation above is done under the valid assumption that $\beta\hbar\omega \ll 1$. This is equivalent to saying $\hbar\omega/k \ll T$

(5) Consider a crystal with N_0 atoms, which, for small oscillations, is equivalent to $3N_0$ decoupled oscillators. The mean thermal energy of the crystal $\overline{E}_{crystal}$ is $\overline{E_{qu}}$ or $\overline{E_{cl}}$ summed over all normal modes. We calculate, if the oscillators are treated classically, the specific heat per atom,

First, the mean thermal energy of a single oscillator is $3k$, thus for N_0 atoms each with 3 decoupled oscillators the crystal's total mean thermal energy is $\overline{E}_{crystal} = 3N_0kT$. Then following the definition of specific heat per atom,

$$C_{cl}(T) = \frac{1}{N_0}\frac{\partial \overline{E}_{crystal}}{\partial T} = \frac{1}{N_0}3N_0k = 3k$$

For the quantum case, we similarly take the mean thermal energy, multiply it by $3N_0$, and follow from the definition of specific heat. Take note of the substitution $\theta_E = \hbar\omega/k$

$$\begin{aligned} C_{qu}(T) &= \frac{1}{N_0}\frac{\partial}{\partial T}\left(3N_0\hbar\omega\left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega}-1}\right)\right) \\ &= 3\theta_E k\frac{\partial}{\partial T}\left(\frac{1}{2} + \frac{1}{e^{\theta_E/T}-1}\right) \\ &= 3\theta_E k\left(\frac{-1}{(e^{\theta_E/T}-1)^2}\right)\left(-\frac{-\theta_E}{T^2}e^{\theta/T}\right) \\ &= 3k\frac{\theta_E^2}{T^2}\frac{e^{\theta_E/T}}{(e^{\theta_E/T}-1)^2} \end{aligned}$$

In the classical case the value is the same for all crystals independent of temperature (except for as $T \to 0$. The quantum case can be shown to recover the classical case when the temperature is greater than the Einstein temperature θ_E. Here we expand our exponentials around a small θ_E/T, do some cancellation, and arrive at the desired expression $3k$ provided we ignore terms that are of order $(\theta_E/T)^2$.

$$\begin{aligned} C_{qu} \xrightarrow[T\gg\theta_E]{} 3k\left(\frac{\theta_E}{T}\right)^2\frac{1+\frac{\theta_E}{T}+O(\frac{\theta_E}{T})^2}{(\frac{\theta_E}{T})^2(1+\frac{1}{2}\frac{\theta_E}{T}+O(\frac{\theta_E}{T})^2)^2} \\ 3k\left(\frac{\theta_E}{T}\right)^2(1+O(\theta_E/T)) \to 3k \end{aligned}$$

To observe when $T \ll \theta_E$ we can distribute the denominator of the expression and taking the leading order behavior for the polynomial fraction.

$$C_{qu} \xrightarrow[T\ll\theta_E]{} 3k\left(\frac{\theta_E}{T}\right)^2\frac{e^{\theta_E/T}}{(e^{\theta_E/T})^2 - 2e^{\theta_E/T}+1} \approx 3k\left(\frac{\theta_E}{T}\right)^2 e^{-\theta/T}$$